Med-Self-Preference

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August 2026

Abstract

Benchmarking medical AI has required increasingly larger datasets, making human evaluation untenable. The common approach now is to test AI models on a set of tasks and to use large language models (LLMs) as automated judges to score and compare model performance. However, LLM judges may show self-preference bias, where a model favors responses produced by its own model. This has serious implications in medicine since LLM-as-a-judge is readily used and biased evaluations could affect which models are considered safe.

This study examines LLM-as-a-judge self-preference in two medical settings. We evaluate eight generators and judges on all 620 Real-POCQi questions and six clinician models and judges on 200 MedSP1000 standardized-patient scenarios at two, four, six, and eight visible turns. Model identities are concealed from the judge in the primary experiments and response order is randomized to control for positional bias.

In the completed Real-POCQi experiment, judges ranked their own exact model's answer 1.219 positions higher on average than other judges ranked that same answer. The effect remained after presentation-position adjustment, within-judge score calibration, and removal of the explicit rubric. In MedSP1000, the corresponding effects were 0.518, 0.457, 0.502, and 0.613 positions at two, four, six, and eight turns. Self-preference varied substantially across judges and did not increase monotonically with every additional pair of turns. Clinical specialty did not explain significant heterogeneity in the Real-POCQi effect. These findings show that raw win rate, score-level preference, and matched rank preference capture different aspects of self-preference and should be considered together when evaluating LLM judges in medicine.

1 Introduction

The complexity of medical practice has called for larger and more complex benchmarking for comparing the performance of AI models. Human review of thousands of scenarios tested across multiple AI models is untenable due to cost and time [1]. A recent review of medical AI benchmarking found that large language models (LLMs) are increasingly used as judges to evaluate the outputs of other LLMs, and in some cases their own outputs [2]. This same study found that OpenAI's models are currently the most popular judges for medical AI benchmarking studies [2].

Researchers claim using an LLM as a judge provides a scalable way to evaluate responses based on certain criteria such as accuracy, completeness, safety, and clarity. The stakes in medical AI evaluation are high, as biased judges could incorrectly favor unsafe models and influence clinical deployment decisions to the detriment of patients. Furthermore, evaluating medical responses is especially complicated since an answer may sound confidently correct, but still contain incomplete medical information. As a result, evaluating LLMs used in the medical context is a multifaceted process where faithfulness, completeness, safety, clarity, and conciseness should be considered.

Prior work in general generative AI tasks has shown that LLM judges may exhibit biases such as positional and verbosity bias. Wataoka et al. examined how LLM judges may favor responses that reflect policies or writing styles familiar to them [3]. One study showed that LLM judges may favor AI-generated clinical responses over human-produced ones. Alvarez-Arenas et al. found that this preference seemed to depend more on how an answer was written, especially how long it was and the language used, than on whether the model could actually tell if the answer came from a human or an AI [4]. Panickssery et al. found that LLM evaluators can recognize and favor their own generations on general tasks [6]. Pombal et al. found that self-preference can persist even with fully objective rubrics [5].

The literature for self-preference in medical contexts is far more sparse. To address these gaps, we evaluate self-preference across multiple model families in both single-turn generalist-specialist question answering and multi-turn patient-physician conversations. In order to address the fact that a model may choose its own response because its answer is genuinely better, we hold the answer fixed and compare how its own model ranks it with how every outside judge ranks that exact answer. We also evaluate calibration-aware rubric scores and a no-rubric ranking condition. Our findings provide important considerations for future medical AI benchmarking studies.

2 Methods

2.1 Generator and judge models

The study uses two models from each of four families, shown in Table 1. These tier labels describe their role in this study and are not intended as a universal performance ranking.

Table 1: Generator and judge cohort.

Family	Higher-capability model	Workhorse model
OpenAI	`gpt-5.6-sol`	`gpt-5.6-terra`
Anthropic	`claude-opus-5`	`claude-sonnet-5`
Google	`gemini-3.1-pro-preview`	`gemini-3.7-flash`
Qwen	`Qwen/Qwen3.5-122B-A10B-FP8`	`Qwen/Qwen3.8-27B-FP8`

2.2 Single-turn pipeline

The single-turn pipeline uses all 620 questions from the frozen Real-POCQi source revision. All eight models generated one answer per question, for 4,960 logical generations. In the primary combined condition, all eight models judged every question. Each judge received the question and eight blinded candidate answers in randomized order. It assigned a score from 0 to 5 on accuracy, clinical utility, source quality, verifiability, and completeness, and then independently ranked all eight answers from best to worst.

A no-rubric direct-ranking sensitivity used the same eight judges on a deterministic sample of 100 questions. Judges were asked only to rank the eight blinded candidates by overall quality. An identity-revealed follow-up uses a seeded 200-question sample and the six API judges; this follow-up was still in progress at the analysis cutoff.

2.3 Multi-turn pipeline

The multi-turn experiment uses the role-separated MedSP1000 dataset. A deterministic 200-scenario cohort was selected from source revision `55e3e55efd08c73baab912ba0c5b42637114fbc8`. Each scenario has a private standardized-patient actor packet and a separate clinician-visible initialization. The patient model receives only the actor packet and visible conversation; the clinician receives only its initialization and visible conversation. Evaluator and environment-controller materials are excluded.

The patient simulator is fixed to `mistralai/Mistral-Small-3.1-24B-Instruct-2503`. Six API clinician models have complete production trajectories: Sol, Terra, Opus, Sonnet, Gemini Pro, and Gemini Flash. Each trajectory contains four clinician-patient exchanges. The same trajectories are judged after two, four, six, and eight visible turns. At each length, all six judges score and rank all six blinded candidates, yielding 1,200 judgments per length. The two planned Qwen clinician and judge extensions have not yet been run for MedSP1000.

3 Analysis of Data

3.1 Matched rank self-preference

For question q and exact model j, let r_{kqj} be the rank that judge k assigns to the answer generated by model j, where lower ranks are better. The matched self-preference contrast is

$$D_{jq} = r_{jqj} - \frac{1}{J-1} \sum_{k \neq j} r_{kqj}. \tag{1}$$

Thus, $D_{jq} < 0$ means that judge j ranks its own answer more favorably than outside judges rank the same answer to the same question. Matching on the exact answer removes general generator quality as an explanation.

Candidate order has a measurable association with rank. Ranks are adjusted with an additive model containing judge-by-generator fixed effects and categorical presentation-position effects. For pooled intervals, the correlated judge effects are first averaged within question and uncertainty is then calculated across questions. Calibration-aware rubric sensitivities standardize each five-axis rubric sum within every judge-question candidate list before applying the same matched comparison. Judge-specific families of tests use Holm correction.

Because the four MedSP1000 conditions are truncations of the same 200 trajectories, transcript-length comparisons are repeated measures. The joint length test uses the 4-minus-2, 6-minus-2, and 8-minus-2 question-level contrasts. Individual length contrasts are paired within question.

4 Results

4.1 Real-POCQi generation and aggregate quality

All 4,960 intended answer cells are present. Claude Opus 5 was materially more verbose than the other generators, with a median of 2,927 output tokens, compared with 1,354 for Sonnet and

656–988 for the remaining six models. Manual review of three matched cases found a substantive venous-reflux error, an internal HSV-trial contradiction, and omissions or unsupported claims across models in an oncology question. The generation set is therefore a valid evaluation input, but not a clinical gold standard.

In the completed combined condition, Opus ranked first overall by a wide margin. The Qwen generations ranked last, although the Qwen judges preserved the broad aggregate order.

Table 2: Real-POCQi aggregate generator performance across eight judges. Lower mean rank is better.

Generator	Mean rank	Ranked first	Mean rubric sum
Claude Opus 5	1.887	69.3%	23.110
GPT-5.6 Terra	3.371	17.5%	21.652
Gemini 3.7 Flash	3.919	1.3%	21.079
GPT-5.6 Sol	4.301	8.3%	20.838
Claude Sonnet 5	4.389	1.5%	20.632
Gemini 3.1 Pro	4.620	1.3%	20.414
Qwen 3.5 122B	6.524	0.4%	17.524
Qwen 3.8 27B	6.989	0.3%	16.406

4.2 Real-POCQi self-preference

Every judge's matched estimate is negative. Every judge-specific interval excludes zero after Holm correction. The pooled position-adjusted effect is -1.219 rank positions, with a 95% CI of $[-1.261, -1.177]$. The unadjusted estimate is also -1.219, indicating that position imbalance does not explain the matched effect.

Table 3: Position-adjusted matched self-preference in Real-POCQi. Negative values indicate that the judge ranked its own answer more favorably.

Judge	Rank effect	95% CI
GPT-5.6 Sol	-2.939	$[-3.041, -2.836]$
GPT-5.6 Terra	-2.224	$[-2.325, -2.123]$
Gemini 3.7 Flash	-1.236	$[-1.317, -1.155]$
Claude Opus 5	-0.969	$[-1.019, -0.919]$
Qwen 3.5 122B	-0.730	$[-0.852, -0.608]$
Gemini 3.1 Pro	-0.683	$[-0.790, -0.575]$
Claude Sonnet 5	-0.647	$[-0.763, -0.530]$
Qwen 3.8 27B	-0.324	$[-0.419, -0.228]$
Pooled within question	-1.219	$[-1.261, -1.177]$

The Qwen results show why raw first-place rate is insufficient. Qwen answers remain weak in absolute terms, but each Qwen judge moves its own answer upward relative to how the outside judges rank that same answer. A calibration-aware rubric sensitivity gives a pooled effect of -0.440 within-list standard deviation units (95% CI $[-0.460, -0.420]$), with all eight judge-specific effects negative.

4.3 No-rubric direct-ranking sensitivity

On the deterministic 100-question sample, the pooled matched effect is -1.263 positions (95% CI $[-1.361, -1.164]$). Seven judge-specific intervals exclude zero; Qwen 27B is the exception at -0.155 positions (95% CI $[-0.346, 0.037]$). Direct and rubric-condition rankings agree on 91.5% of candidate pairs, select the same first-place answer in 87.1% of cells, and produce the identical complete order in 16.1%. Removing the explicit rubric therefore changes fine ordering but does not remove the main self-preference pattern.

4.4 Clinical specialty analysis

All adequately sized specialties have negative point estimates. Among specialties with at least ten questions, the pooled effects range from -1.543 positions in Family Medicine to -0.908 in Radiology. However, a label-permutation test preserving the 30 observed specialty sizes finds that specialty explains only 5.7% of question-level variation and does not show significant global heterogeneity ($p = 0.190$). No judge-specific specialty test crosses 0.05. The correct interpretation is not that every specialty has an identical true effect, but that the current data do not provide convincing evidence of a specialty interaction.

4.5 MedSP1000 results

All four transcript-length conditions are complete at 1,200 judgments each. Matched self-preference is present at every length. The calibration-aware rubric sensitivity points in the same direction.

Table 4: Pooled MedSP1000 matched self-preference by visible transcript length.

Visible turns	Rank effect	95% CI	Normalized-rubric effect
2	-0.518	$[-0.585, -0.452]$	-0.309
4	-0.457	$[-0.527, -0.388]$	-0.305
6	-0.502	$[-0.574, -0.430]$	-0.356
8	-0.613	$[-0.683, -0.543]$	-0.405

Judge-specific patterns vary over length and do not share a simple trajectory. Opus becomes more self-preferring with length, Gemini Pro is strongest at two turns, and Sol has a marked increase only at eight turns.

Table 5: Judge-specific MedSP1000 matched rank effects.

Judge	2 turns	4 turns	6 turns	8 turns
Claude Opus 5	-0.153	-0.218	-0.423	-0.445
Claude Sonnet 5	-0.233	-0.378	-0.549	-0.472
Gemini 3.1 Pro	-0.791	-0.554	-0.436	-0.545
Gemini 3.7 Flash	-0.572	-0.600	-0.428	-0.645
GPT-5.6 Sol	-0.536	-0.487	-0.539	-0.938
GPT-5.6 Terra	-0.826	-0.508	-0.637	-0.632

A repeated-measures test rejects equality across the four lengths ($F(3, 197) = 3.972$, $p = 0.0089$). The 4-minus-2 and 6-minus-2 changes are small and do not exclude zero. The 8-minus-2 change is -0.094 positions (95% CI $[-0.187, -0.002]$, nominal $p = 0.045$), but does not survive Holm correction across the three baseline contrasts. The adjacent 8-minus-6 change is -0.111 positions

(95% CI $[-0.189, -0.033]$, nominal $p = 0.0054$). The defensible conclusion is that self-preference varies with visible length and is strongest at eight turns in the pooled data, not that it rises monotonically from two through eight turns.

4.6 Experiments still in progress or not completed

The identity-revealed Real-POCQi experiment was actively running at the fixed August 24 snapshot. It had 789 of 1,200 planned latest successful judgments, with 128 of 200 questions complete for all six API judges. Interim results are not treated as final in the main conclusions. The MedSP1000 Qwen expansion, production direct-ranking condition, and production rubric-sum-only condition have not been run. Physician validation of the judging conditions also remains incomplete.

5 Discussion

As medical benchmarking and monitoring relies more on LLMs-as-a-judge, it has become increasingly important to understand how LLMs exhibit self-preference. Our study focuses on healthcare, where benchmarking results can impact which models are selected for implementation. We examine self-preference across different model families, separating scoring generosity from direct preferences and from a matched comparison in which the exact answer is held fixed.

The experiments show that self-preference can look different depending on how it is measured. Every Real-POCQi judge ranks its own answer more favorably than outside judges rank the same answer, including models whose answers rarely rank first. This effect remains after position adjustment, within-judge score calibration, and removal of the explicit rubric. It is safer to describe this as same-model affinity rather than conscious self-recognition. Generator identities were hidden, so the effect may arise from shared style, reasoning conventions, favored evidence, or family-specific quality criteria.

The multi-turn findings qualify the role of conversation length. Self-preference is present at every tested MedSP1000 length and is strongest at eight turns in the pooled data, but it does not steadily increase from two to four to six to eight turns. Different judges show different trajectories. Single-turn benchmarks may still miss behavior that appears in longer conversations, but the length effect should not be described as universally monotonic.

Clinical specialty does not explain reliable heterogeneity in the current Real-POCQi sample. The magnitude is driven much more by which model is judging. This suggests that medical benchmarking studies should report results by judge and use multiple judge families rather than relying on one pooled evaluator.

5.1 Limitations

There are limitations to our study. The experiments have one saved generation per model-question cell and one successful judging realization per logical cell. The analysis therefore cannot separate stable model behavior from generation-specific or run-to-run idiosyncrasy.

Rankings are mutually dependent within each candidate list. Question-level intervals address clustering for the pooled estimand, but remain large-sample approximations. Exact-model and broader family affinity are also difficult to separate completely. Response length and stylistic differences may mediate the effect. Opus is especially verbose in Real-POCQi, and clinician output length also differs in MedSP1000.

The specialty labels are broad and highly imbalanced, ranging from three to 55 questions. The absence of detected heterogeneity is not proof that narrower clinical domains have identical effects. The identity-revealed experiment is in progress, the full MedSP1000 Qwen extension has not been run, and the automated judges have not yet been benchmarked against completed blinded physician judgments. Finally, the analysis was exploratory and was not preregistered. Effect sizes and sensitivity checks should be emphasized over isolated p-values.

6 Conclusion

Overall, our findings show that self-preference is present in medical LLM judging, but it does not appear in the same way across all models or metrics. The experiments show that LLM judges rank answers from their own exact model more favorably than other judges rank the same answers. This pattern is robust across two medical settings, position adjustment, calibrated rubric scores, and a no-rubric sensitivity analysis. In multi-turn conversations it is present at every tested length and strongest at eight turns in pooled data, without a monotonic increase from two to eight turns. These results highlight the importance of using multiple judge families, multiple evaluation measures, and matched analyses when comparing medical LLMs.

A Complete experiment status

Table 6: Status of experiments included in the project at the fixed analysis snapshot.

Phase	Experiment	Status
Current	Eight-model Real-POCQi generation, 4,960 answers	Complete
Current	Blinded Real-POCQi combined judging, 4,960 judgments	Complete
Current	Real-POCQi no-rubric direct ranking, 800 judgments	Complete
Current	Real-POCQi specialty analysis	Complete
Current	Identity-revealed Real-POCQi, 1,200 planned judgments	**In progress**; 789 successes at snapshot
Current	Six-model MedSP1000 generation, 1,200 trajectories	Complete
Current	MedSP1000 judging at 2, 4, 6, and 8 turns, 1,200 each	Complete
Planned	MedSP1000 Qwen clinician and judge extension	Not started
Planned	MedSP1000 production direct ranking and rubric-sum-only conditions	Pilot only
Validation	Blinded physician judging benchmark	Not completed

B Real-POCQi specialty estimates

Table 7: Position-adjusted matched self-preference by clinical specialty. Negative values indicate self-preference.

Specialty	Questions	Rank effect	Rubric-rank sensitivity
Endocrinology	55	-1.359	-1.320
Cardiology	54	-1.189	-1.064
Oncology / Hematology	49	-1.052	-1.029
OB / GYN	44	-1.283	-1.216
Dermatology	40	-1.097	-1.113
Rheumatology	39	-1.234	-1.166
Neurology	32	-1.231	-1.158
Infectious Disease	30	-1.322	-1.348
Psychiatry	25	-1.359	-1.244
Surgery	25	-1.371	-1.225
Nephrology	20	-1.172	-1.048
Allergy & Immunology	19	-1.249	-1.128
Pediatrics	19	-1.188	-1.193

Specialty	Questions	Rank effect	Rubric-rank sensitivity
Internal Medicine	17	-1.256	-1.235
Critical Care	14	-1.221	-1.197
Gastroenterology	14	-1.141	-1.101
Pulmonology	14	-1.197	-1.065
Transplantation	14	-1.140	-1.044
Anesthesiology	11	-1.297	-1.091
Family Medicine	11	-1.543	-1.308
Orthopedics	11	-1.215	-1.265
Primary Care	11	-1.397	-1.258
Emergency Medicine	10	-1.243	-1.232
Radiology	10	-0.908	-1.023
Urology	8	-1.119	-0.897
Hospital-Based Medicine	7	-0.862	-0.684
Surgical Oncology	6	-1.063	-0.913
Genetics	4	-0.684	-0.645
Otolaryngology	4	-0.933	-0.799
Ophthalmology	3	-1.056	-0.997

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